



DATA STRUCTURES

(Course Code: 25B11CI318)

Overview

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What is a Data Structure?

Definition

A **data structure** is a logical or mathematical model that defines how data is organized, stored, and related, along with the operations that can be performed on it to achieve efficient computation.

Formal Definition (Horowitz & Sahni)

A data structure is a collection of data elements whose organization is characterized by the relationships among the data elements and the operations that can be performed on the data.

Mathematical Representation

A data structure d is represented as

$$d = (D, F, A)$$

where

D – Data

A finite set of data elements (domain of values).

F – Mapping Functions

A set of functions (relationships) that specify how the data elements are organized.

A – Admissible Operations

A set of operations (algorithms) that manipulate the data.

- Insert, Delete,
- Search,
- Traverse
- Update

Example: Linked List

Consider the linked list

10 $\longrightarrow$ 20 $\longrightarrow$ 30 $\longrightarrow$ NULL

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Data (D):

Example: Linked List

Consider the linked list

$10 \longrightarrow 20 \longrightarrow 30 \longrightarrow \text{NULL}$

Data (D): $\{10, 20, 30\}$

Mapping Functions (F):

Example: Linked List

Consider the linked list

$10 \longrightarrow 20 \longrightarrow 30 \longrightarrow \text{NULL}$

Data (D): $\{10, 20, 30\}$

Mapping Functions (F): $10 \rightarrow 20 \rightarrow 30 \rightarrow \text{NULL}$

Admissible Operations (A):

Example: Linked List

Consider the linked list

$10 \longrightarrow 20 \longrightarrow 30 \longrightarrow \text{NULL}$

Data (D): $\{10, 20, 30\}$

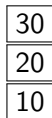
Mapping Functions (F): $10 \rightarrow 20 \rightarrow 30 \rightarrow \text{NULL}$

Admissible Operations (A): $\{\text{Insert, Delete, Search, Traverse, Update}\}$

$LinkedList = (D, F, A)$

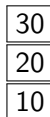
Example: Stack

Consider the following stack:



Example: Stack

Consider the following stack:



Data (D): $\{10, 20, 30\}$

Mapping Functions (F): Elements are organized as **LIFO** (Last-In, First-Out) principle.

Admissible Operations (A): $\{\text{Push, Pop, Peek, IsEmpty, IsFull}\}$

$$\boxed{Stack = (D, F, A)}$$

Classification of Data Structures

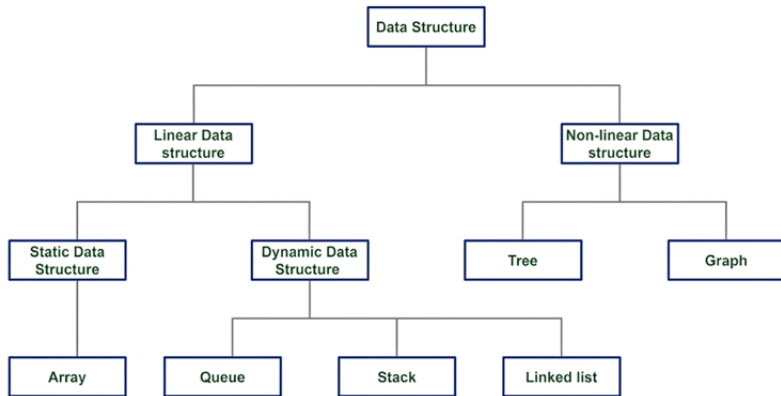


Figure: Classification of Data Structures

Classification of Data Structures

Data structures are broadly classified into two categories:

① Linear Data Structures

- Elements are arranged sequentially.
- Each element (except the first and last) has a unique predecessor and successor.

② Non-linear Data Structures

- Elements are organized hierarchically or as a network.
- One element can be connected to multiple elements.

Linear Data Structures

A **Linear Data Structure** stores elements in a sequential order.

Types

- **Static Data Structure**

- Memory size is fixed.
- Example: **Array**

- **Dynamic Data Structure**

- Memory size can grow or shrink during execution.
- Examples:
 - Linked List
 - Stack
 - Queue

Non-linear Data Structures

A **Non-linear Data Structure** organizes data in a hierarchical or network form.

Examples include:

- **Tree**

- Hierarchical organization.
- Used in file systems, databases, expression trees.

- **Graph**

- Collection of vertices connected by edges.
- Used in social networks, maps, routing algorithms.